YOUSEF ALSAYED ABDELFATTAH

Cairo, Egypt • yousefabuelkhear@gmail.com • +20 115 382 6643 • linkedin.com/in/yousefalsayed-abdelfattah • github.com/yousefattaff

EDUCATION

B.Sc. Mechatronics, Robotics & Automation Engineering | Misr University for Science & Technology (MUST), Giza Sep 2019 – Jun 2024

- Graduation Project: R2-Waiter — autonomous restaurant waiter robot with ROS , Nav, PyQt5 order management, and animated real-time emotion display.

EXPERIENCE

AI Software Engineer | *Outlier AI, Remote* Nov 2024 – Jan 2025

- Evaluated and ranked LLM outputs across diverse prompts to improve model accuracy and alignment.
- Identified systematic failure modes and provided structured feedback to enhance prompt–response quality.

PLC Automation Engineer (Intern) | *Siemens, Cairo* Sep 2023

- Designed Ladder Logic automation programs in TIA Portal for S7-1200/1500 PLCs; debugged live factory workflows.

CNC Programming & Operation (Intern) | *DMG MORI, Cairo* Sep 2022

- Developed G-code programs for turning and milling; operated CNC machines to precision engineering tolerances.

PROJECTS

SDR Agent — *OpenAI Agents SDK, Python, Groq, Gemini, SendGrid*

- End-to-end Sales Development Representative pipeline: research planner spawns three parallel agents to build a prospect summary, then orchestrates three LLM drafters (Llama 3.3 70B / Gemini 2.0 Flash / gpt-oss-120b) generating competing cold emails.
- Sales manager agent selects the best draft via Pydantic structured output; input guardrail blocks personal names; final email formatted as HTML and sent via SendGrid with full OpenAI Traces observability.

RAG AI Agent — *LangChain, ChromaDB, Hugging Face, Groq, Streamlit, Edge TTS*

- Streamlit app for building named knowledge-base experts from PDFs; each expert maps to a ChromaDB collection with semantic retrieval and in-UI source citations (filename + page).
- Agentic mode equips the LLM with PubMed, Tavily web search, and clinical calculators (BMI, eGFR, BSA); supports cloud (Groq/Llama 3.3 70B) and fully offline (Ollama) backends.

EgyptianSavoryScout — *QWEN 1.5B (LoRA/PEFT), TF-IDF, FastAPI, React/TanStack*

- Fine-tuned QWEN 3.5 1.5B on domain-specific data for restaurant recommendation across 1,000+ real Egyptian venues in 8 cities; TF-IDF ML ranker applies hard city filtering + cosine similarity before LLM inference.
- FastAPI backend streams ranked candidates and final recommendation to a React/TanStack frontend with city keyword detection.

R2-Waiter (Graduation Project) — *ROS 2 , Nav2, PyQt5, Python*

- PyQt5 desktop controller for TurtleBot3 Waffle: 3-slot tray management, 12-table order grid, and Nav2 sequential goal dispatch with odometry feedback and auto-recovery.
- Animated HTML robot face (12 expressions via PyQtWebEngine) tracks /cmd_vel turn direction in real time.

AI Engineering Multi-Agent Crew — *CrewAI, LangChain, Python*

- Three-agent software crew (Architect, Developer, Reviewer) using CrewAI orchestration; **Orchestrated** a three-agent software crew (Architect, Developer, Reviewer) using CrewAI with sequential and hierarchical process flows, assigning each agent a specialized free LLM: Qwen3 32B (Groq) for architecture planning, openai/gpt-oss-120b (Groq) for code generation, and moonshotai/kimi-k2.6:free (OpenRouter) for code review.
- **Configured** multi-provider routing via a single .env to stay entirely within free tiers — Groq for ultra-low-latency inference (500+ tokens/sec) and OpenRouter as fallback — enabling zero-cost end-to-end software delivery from spec to reviewed code.

LLMs & AI: RAG Pipelines, AI Agents, Prompt Engineering, Fine-tuning (LoRA/PEFT), Embeddings & Semantic Search

Frameworks: LangChain, LangGraph, CrewAI, OpenAI Agents SDK, FastAPI, Streamlit, Pydantic

Tools & Languages: Python, C, Git, GitHub, Docker, ChromaDB, Sentence-Transformers, ROS 2

CERTIFICATIONS

AI Engineer Core Track (Udemy) • Complete Agentic AI Engineering — Ed Donner (2025) • CS50x & CS50 AI — Harvard edX • AI For Everyone — DeepLearning.AI • Embedded Systems AVR — IMT • EF SET C2 English